Spatial Extension of the Geometric Matrix

Rational Foundations for Modular 3D Systems

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Abstract

This paper extends the Geometric Matrix framework into three-dimensional space. Planar interval states $[a, d]$ are characterized by the structural invariants centroid b, span c, offset x, altitude y, and the Delta invariant Δ, which classify configurations as Equilibrium Right ($\Delta = 0$), Obtuse ($\Delta < 0$), or Acute ($\Delta > 0$).

The three-dimensional extension embeds the planar rectangular envelope $A_{\text{envelope}} = cy$ into space through an orthogonal depth parameter $z = 2k^2$, where k is a discrete grid parameter. This yields the triangular-prism volume

$$V = cyk^2,$$

exactly equivalent to the classical Euclidean prism volume $V_{\text{standard}} = A_{\text{planar}} \cdot z$. The grid parameter is constructed directly from the planar invariants as $k = cyd/b$. For the Equilibrium Right state $[1, 5]$, this construction produces integer altitude ($y = 3$) and integer k ($k = 20$), yielding the exact volume $V = 4{,}800$. The Obtuse and Acute representative states produce the same volume expression but with irrational altitude and grid values. The resulting state vector $P = (b, c, k)$ provides a compact representation linking planar Matrix geometry to its three-dimensional spatial envelope.

1 Planar Foundations and Structural Invariants

Classical geometry commonly represents spatial configurations through external Cartesian coordinates, vertex arrays, or derived trigonometric quantities. The Geometric Matrix framework instead parametrizes planar configurations directly from a primitive closed interval state

$$[a, d] \subset \mathbb{R}, \qquad a < d.$$

The interval establishes the primary boundary from which the remaining geometric quantities are derived.

1.1 Fundamental Metric Definitions

For any interval state $[a, d]$, define the following primary invariants.

Left Anchor. The lower scalar boundary of the interval state is denoted a.

Centroid. The arithmetic midpoint of the interval is

$$b = \frac{a+d}{2}.$$

Span. The linear extent of the interval is

$$c = d - a.$$

Right Boundary. The upper scalar boundary of the interval state is denoted d.

Thus, the planar Matrix state is generated from the pair $[a, d]$, with b and c determined directly from those boundaries. This terminology is consistent with the foundational treatment in Johnson (2026) [1].

1.2 The Planar Delta Invariant

The geometry of the embedded planar state is governed by the structural invariant

$$\Delta = b^2 + c^2 - d^2.$$

Substituting $b = \frac{a+d}{2}$ and $c = d - a$:

$$\Delta = \left(\frac{a+d}{2}\right)^2 + (d-a)^2 - d^2.$$

Expanding:

$$\Delta = \frac{a^2 + 2ad + d^2}{4} + d^2 - 2ad + a^2 - d^2,$$

and therefore

$$\Delta = \frac{5a^2 - 6ad + d^2}{4} = \frac{(5a-d)(a-d)}{4}.$$

Because $a < d$, the factor $(a - d)$ is negative. Consequently, the sign of Δ is controlled by the relationship between d and $5a$.

1.3 Geometric Projection Parameters

The planar state is embedded through two orthogonal projection components. Both quantities appear in Johnson (2026) [1] as intermediate coordinates of the triangle's third vertex, derived within the circumcenter construction. Here they are treated as primary planar invariants in their own right, since the three-dimensional extension depends on them directly.

Horizontal Offset. The structural displacement of the vertex from the interval midpoint is

$$x = \frac{\Delta}{2c}.$$

Planar Altitude. The orthogonal height of the embedded planar state is

$$y = \sqrt{b^2 - x^2}.$$

Thus the relation $x^2 + y^2 = b^2$ provides the local Euclidean decomposition of the centroid radius into orthogonal planar components.

Planar Envelope Area. The rectangular envelope containing the embedded planar state is defined by the span c and altitude y:

$$A_{\text{envelope}} = cy.$$

The corresponding Euclidean area of the embedded triangle is one-half of this envelope:

$$A_{\text{planar}} = \frac{1}{2}cy = \frac{1}{2}A_{\text{envelope}}.$$

2 Family Classification via the Delta Invariant

The sign of $\Delta = \dfrac{(5a-d)(a-d)}{4}$ partitions the planar interval states into three structural families.

$\Delta = 0$	$\Delta < 0$	$\Delta > 0$
Equilibrium Right	Obtuse	Acute
$d = 5a$	$d < 5a$	$d > 5a$

2.1 Equilibrium Right Family ($\Delta = 0$)

The condition $\Delta = 0$ requires $d = 5a$. Since $x = \Delta/2c$, the offset vanishes: $x = 0$. The altitude therefore becomes $y = b$. The resulting planar geometry satisfies $b^2 + c^2 = d^2$, so the embedded triangle is a right triangle.

For the normalized state $[a, d] = [1, 5]$, the resulting parameters are $b = 3$, $c = 4$, $d = 5$, giving the canonical 3-4-5 configuration.

2.2 Obtuse Family ($\Delta < 0$)

For $\Delta < 0$, the condition is $d < 5a$. The offset satisfies $x < 0$, so the vertex is displaced in the negative direction relative to the interval midpoint. The resulting embedded configuration is classified as obtuse.

The representative normalized state $[a, d] = [1, 3]$ has $\Delta = -1$.

2.3 Acute Family ($\Delta > 0$)

For $\Delta > 0$, the condition is $d > 5a$. The offset satisfies $x > 0$, so the vertex is displaced in the positive direction relative to the interval midpoint. The resulting embedded configuration is classified as acute.

The representative normalized state $[a, d] = [1, 9]$ has $\Delta = 8$.

3 Comparative Analysis of Representative Planar States

To establish numerical and structural baseline properties across all three families, we evaluate three normalized interval states with $a = 1$: $[1, 3]$, $[1, 5]$, $[1, 9]$, representing the Obtuse, Equilibrium Right, and Acute families respectively.

Parameter / Invariant	Obtuse $[1, 3]$	Equilibrium Right $[1, 5]$	Acute $[1, 9]$
Bound a	1	1	1
Bound d	3	5	9
Centroid b	2	3	5
Span c	2	4	8
Delta Δ	-1	0	$+8$
Offset x	$-\frac{1}{4}$	0	$\frac{1}{2}$
Altitude y	$\frac{3\sqrt{7}}{4}$	3	$\frac{3\sqrt{11}}{2}$
Envelope Area cy	$\frac{3\sqrt{7}}{2}$	12	$12\sqrt{11}$
Triangle Area $\frac{1}{2}cy$	$\frac{3\sqrt{7}}{4}$	6	$6\sqrt{11}$

The three states exhibit the same underlying Matrix construction while differing in the arithmetic character of their derived geometric quantities. The Equilibrium Right state is the only one of these representatives for which the altitude is an integer.

4 Volumetric Derivations and Spatial Calibration

4.1 Classical Grounding and Envelope Extrusion

All three-dimensional volumetric operations in the Matrix framework are defined using standard Euclidean spatial measures. The framework does not introduce an alternative definition of volume; it introduces a deterministic parameterization of a classical spatial envelope.

For any planar interval state $[a, d]$, the rectangular envelope area is $A_{\text{envelope}} = cy$, and the embedded triangle has classical Euclidean area $A_{\text{planar}} = \frac{1}{2}cy$.

Extruding the planar triangle along an orthogonal axis to a depth z produces the standard right-prism volume

$$V_{\text{standard}} = A_{\text{planar}} \cdot z = \frac{1}{2}cy \cdot z.$$

The Matrix construction introduces a discrete grid parameter $k \in \mathbb{N}$ through the reparameterization

$$z = 2k^2.$$

Substituting into the classical prism equation:

$$V_{\text{standard}} = \frac{1}{2}cy(2k^2),$$

so

$$V(k) = cyk^2.$$

The Matrix constructive volume is therefore exactly the same Euclidean prism volume, expressed through the discrete parameter k.

4.2 The Matrix Grid Parameter

The volume $V(k) = cyk^2$ is defined for any admissible $k \in \mathbb{N}$; nothing in Section 4.1 privileges one value of k over another. To connect the depth parameter to the planar state itself, rather than treating it as a free external choice, we define the **Matrix Grid Parameter** directly from the four primary planar invariants:

$$k = \frac{cyd}{b}.$$

This is a direct construction, not a value solved for by equating $V(k)$ to any second, independently motivated volume expression. It combines the envelope's linear dimensions (c, y) with the interval's boundary ratio (d/b) into a single length-scale intrinsic to the planar state.

Whether k is rational, and whether it lies in $\mathbb{N}$, depends entirely on the arithmetic character of the state $[a, d]$ from which it is built. Section 4.3 evaluates this directly across the three representative families.

4.3 Worked Examples Across Matrix Families

We evaluate $k = cyd/b$ and the resulting volume $V = cyk^2$ across all three representative families.

Case 1: Equilibrium Right State ($\Delta = 0$).

State: $[a, d] = [1, 5]$. Parameters: $b = 3$, $c = 4$, $d = 5$, $\Delta = 0$, $x = 0$, $y = 3$.

Envelope area: $A_{\text{envelope}} = cy = 4(3) = 12$. Triangle area: $A_{\text{planar}} = 6$.

Grid parameter:

$$k = \frac{cyd}{b} = \frac{(4)(3)(5)}{3} = 20.$$

Since $k \in \mathbb{N}$, this state produces an integer grid value.

Volume:

$$V = cyk^2 = 12(20)^2 = 12(400) = 4{,}800.$$

Classical check. With $z = 2k^2 = 800$:

$$V_{\text{standard}} = A_{\text{planar}} \cdot z = 6(800) = 4{,}800.$$

The two expressions agree, as guaranteed by the identity established in Section 4.1 — this is a consistency check of the extrusion algebra, not an independent verification of k itself.

Case 2: Obtuse State ($\Delta < 0$).

State: $[a, d] = [1, 3]$. Parameters: $b = 2$, $c = 2$, $d = 3$, $\Delta = -1$, $x = -\frac{1}{4}$.

Altitude: $y = \sqrt{2^2 - (-\frac{1}{4})^2} = \sqrt{\frac{63}{16}} = \dfrac{3\sqrt{7}}{4}$.

Envelope area: $A_{\text{envelope}} = 2\left(\dfrac{3\sqrt{7}}{4}\right) = \dfrac{3\sqrt{7}}{2}$. Triangle area: $A_{\text{planar}} = \dfrac{3\sqrt{7}}{4}$.

Grid parameter:

$$k = \frac{cyd}{b} = \frac{2\left(\frac{3\sqrt{7}}{4}\right)(3)}{2} = \frac{9\sqrt{7}}{4} \approx 5.953.$$

$k \notin \mathbb{Q}$.

Volume:

$$V = cyk^2 = \frac{3\sqrt{7}}{2}\left(\frac{9\sqrt{7}}{4}\right)^2 = \frac{3\sqrt{7}}{2} \cdot \frac{567}{16} = \frac{1701\sqrt{7}}{32} \approx 140.638.$$

Classical check. With $z = 2k^2 = \dfrac{567}{8}$:

$$V_{\text{standard}} = \frac{3\sqrt{7}}{4} \cdot \frac{567}{8} = \frac{1701\sqrt{7}}{32}.$$

Agreement confirmed.

Case 3: Acute State ($\Delta > 0$).

State: $[a, d] = [1, 9]$. Parameters: $b = 5$, $c = 8$, $d = 9$, $\Delta = 8$, $x = \frac{1}{2}$.

Altitude: $y = \sqrt{5^2 - (\frac{1}{2})^2} = \sqrt{\frac{99}{4}} = \dfrac{3\sqrt{11}}{2}$.

Envelope area: $A_{\text{envelope}} = 8\left(\dfrac{3\sqrt{11}}{2}\right) = 12\sqrt{11}$. Triangle area: $A_{\text{planar}} = 6\sqrt{11}$.

Grid parameter:

$$k = \frac{cyd}{b} = \frac{8\left(\frac{3\sqrt{11}}{2}\right)(9)}{5} = \frac{108\sqrt{11}}{5} \approx 71.639.$$

$k \notin \mathbb{Q}$.

Volume:

$$V = cyk^2 = 12\sqrt{11}\left(\frac{108\sqrt{11}}{5}\right)^2 = 12\sqrt{11} \cdot \frac{128{,}304}{25} = \frac{1{,}539{,}648\sqrt{11}}{25} \approx 204{,}257.389.$$

Classical check. With $z = 2k^2 = \dfrac{256{,}608}{25}$:

$$V_{\text{standard}} = 6\sqrt{11} \cdot \frac{256{,}608}{25} = \frac{1{,}539{,}648\sqrt{11}}{25}.$$

Agreement confirmed.

5 Unified 3D State Representation and Spatial Geometry

5.1 Minimal 3D State Vector

The planar Matrix geometry is generated from the interval state $[a, d]$, but the primary planar quantities can be reconstructed from the pair (b, c). From $b = (a + d)/2$ and $c = d - a$:

$$d = b + \frac{c}{2}, \qquad a = b - \frac{c}{2}.$$

Thus b and c uniquely determine the original interval boundaries. The three-dimensional extension adds the independent spatial parameter k. The resulting minimal spatial state vector is

$$P = (b, c, k).$$

Given P, the planar interval is reconstructed by $a = b - c/2$, $d = b + c/2$, while $\Delta = b^2 + c^2 - d^2$, $x = \Delta/2c$, $y = \sqrt{b^2 - x^2}$, and $A_{\text{envelope}} = cy$ follow directly. The corresponding three-dimensional Euclidean depth is $z = 2k^2$.

Therefore, the state vector $P = (b, c, k)$ determines the planar Matrix state together with its volumetric spatial extension, without requiring redundant storage of derived quantities.

5.2 Space Diagonal

The rectangular volume envelope has orthogonal dimensions c, y, z. Its Euclidean space diagonal satisfies

$$D_{3D}^2 = c^2 + y^2 + z^2.$$

Because $z = 2k^2$:

$$D_{3D}^2 = c^2 + y^2 + (2k^2)^2 = c^2 + y^2 + 4k^4.$$

Using $y^2 = b^2 - x^2$:

$$D_{3D}^2 = b^2 + c^2 - x^2 + 4k^4.$$

For the Equilibrium Right family, $\Delta = 0 \Rightarrow x = 0$ and $b^2 + c^2 = d^2$, so

$$D_{3D}^2 = d^2 + 4k^4.$$

For the canonical state $(b, c, d, k) = (3, 4, 5, 20)$: $z = 2(20)^2 = 800$, so

$$D_{3D}^2 = 3^2 + 4^2 + 800^2 = 25 + 640{,}000 = 640{,}025,$$

$$D_{3D} = 5\sqrt{25{,}601}.$$

This is the Euclidean diagonal of the actual rectangular volume envelope associated with the $k = 20$ construction.

6 Implementation for Modular Spatial Systems

The mathematical construction provides a compact representation suitable for deterministic spatial indexing and modular spatial computation.

6.1 Unified Data Representation

The state vector $P = (b, c, k)$ provides the primary spatial representation. From b and c, the original interval boundaries are recovered: $a = b - c/2$, $d = b + c/2$. The remaining planar quantities follow deterministically: $\Delta = b^2 + c^2 - d^2$, $x = \Delta/2c$, $y = \sqrt{b^2 - x^2}$. The planar envelope is $A_{\text{envelope}} = cy$, and the Euclidean spatial depth corresponding to k is $z = 2k^2$. The resulting volume is $V = A_{\text{envelope}} k^2 = cyk^2$.

Thus the state vector provides a compact representation from which the interval boundaries, planar projection, envelope, spatial depth, and volume can be reconstructed.

6.2 Deterministic Arithmetic

Where Matrix quantities are rational, exact symbolic or rational arithmetic can be retained through intermediate stages rather than converting immediately to floating-point representations.

For the Equilibrium Right state $(a, d) = (1, 5)$, all principal quantities remain integer-valued:

$$b = 3,\ c = 4,\ d = 5,\ x = 0,\ y = 3,\ A_{\text{envelope}} = 12,\ k = 20,\ V = 4{,}800.$$

The non-equilibrium representative states demonstrate the corresponding appearance of irrational factors through the planar altitude and the grid parameter k itself. This distinction provides a direct arithmetic classification of the resulting spatial states.

7 Comparative Synthesis and Equilibrium Selection

The three representative configurations are summarized below.

Matrix Family	Interval $[a, d]$	Δ	Altitude y	Grid Param. k	Volume V
Equilibrium Right	$[1, 5]$	0	3	20 (Int.)	4,800
Obtuse	$[1, 3]$	-1	$\frac{3\sqrt{7}}{4}$	$\frac{9\sqrt{7}}{4}$ (Irr.)	$\frac{1701\sqrt{7}}{32}$
Acute	$[1, 9]$	$+8$	$\frac{3\sqrt{11}}{2}$	$\frac{108\sqrt{11}}{5}$ (Irr.)	$\frac{1{,}539{,}648\sqrt{11}}{25}$

7.1 Euclidean Admissibility

The constructive volume $V(k) = A_{\text{envelope}} k^2$ maps exactly to the classical right-prism volume through $z = 2k^2$. Since $A_{\text{planar}} = \frac{1}{2} A_{\text{envelope}}$:

$$A_{\text{planar}} \cdot z = \frac{1}{2} A_{\text{envelope}}(2k^2) = A_{\text{envelope}} k^2.$$

Thus, for the three-dimensional configurations evaluated here, the Matrix construction is a deterministic parameterization of the corresponding Euclidean prism volume rather than an alternative volume definition. This holds identically for any admissible $[a, d]$ and any k, independent of the particular value chosen — it follows directly from the algebra in Section 4.1, not from the three worked examples, which serve only as numerical illustration.

7.2 Equilibrium Selection

The Equilibrium Right state is characterized by $\Delta = 0$, which forces $x = 0$ and $y = b$. For the representative state $[1, 5]$, $y = 3$ and $k = cyd/b = 20 \in \mathbb{N}$.

By contrast, the representative Obtuse and Acute states produce irrational values of y and consequently irrational values of k, since k is built directly from y.

Therefore, among the representative states evaluated in this paper, the Equilibrium Right configuration is uniquely distinguished by preservation of integer arithmetic through the grid parameter construction. This is a direct consequence of $\Delta = 0$ forcing rational, indeed integer, planar quantities for the normalized case; it is not a separately derived or privileged property of k itself.

8 Conclusion

The Spatial Extension of the Geometric Matrix establishes a direct mathematical relationship between a planar interval state and its three-dimensional Euclidean spatial envelope.

Beginning with the primitive interval $[a, d]$, the framework derives the centroid b, span c, Delta invariant Δ, offset x, and altitude y. These quantities define the planar rectangular envelope $A_{\text{envelope}} = cy$.

The three-dimensional extension introduces the discrete grid parameter k through the depth reparameterization $z = 2k^2$, yielding the constructive volume

$$V(k) = cyk^2.$$

Because $A_{\text{planar}} = \frac{1}{2}cy$, this construction is exactly equivalent to the classical Euclidean prism volume $V_{\text{standard}} = A_{\text{planar}} \cdot z$, as shown in Section 4.1.

The grid parameter is constructed directly from the planar invariants as $k = cyd/b$, combining the envelope dimensions with the boundary ratio d/b into a single intrinsic length-scale. Evaluated across the representative Equilibrium Right, Obtuse, and Acute states, this construction produces integer output only for the Equilibrium Right family: for $[1, 5]$, $k = 20$ and $V = 4{,}800$ exactly. The non-equilibrium representative states produce the same volume expression but with irrational altitude and grid values, a direct consequence of their non-vanishing Delta invariant.

The resulting spatial state vector $P = (b, c, k)$ provides a compact representation from which the underlying interval boundaries, planar geometry, spatial depth, and volumetric envelope can be reconstructed. The framework therefore supplies a unified mathematical representation linking planar Matrix geometry to modular three-dimensional spatial envelopes.

This exact arithmetic is specific to the Equilibrium Right family as generated by this construction, which is exclusively the 3-4-5 family scaled by $a = n$; other Pythagorean triples, such as 5-12-13, arise from different boundary conditions and are not reachable states of this Matrix. The distinction demonstrated here is between the Equilibrium Right family and the Obtuse and Acute families evaluated in Section 4.3, not between different right-triangle shapes.

The three-dimensional envelope is therefore not a separate framework, but the same planar Matrix, read one dimension further out.

References

[1] Johnson, C. *The Geometric Matrix: Rational Derivation of the Pythagorean Triangle*. SemanticDrift, 2026.